Generating and testing flying focus laser pulses with Lasy for PIConGPU simulations

— A Bachelors Defense —

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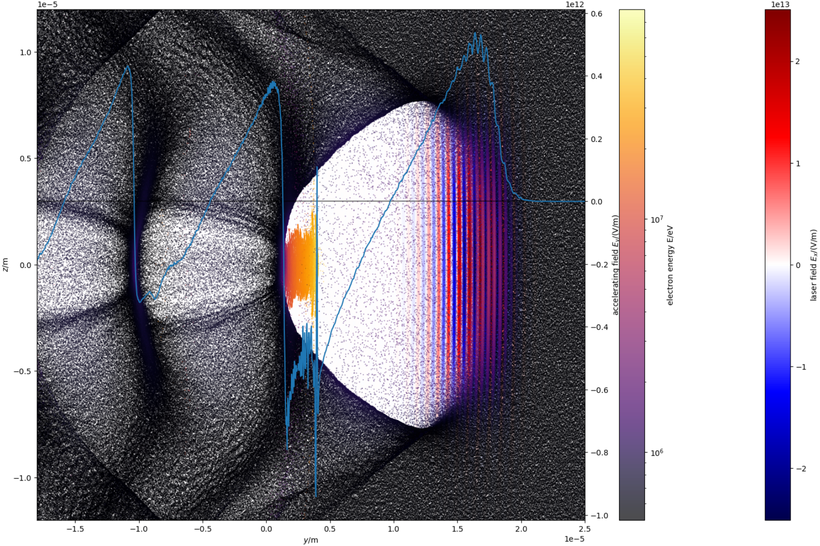
1 Dephasingless Laser WakeField Acceleration (DLWFA)

Laser WakeField Acceleration (LWFA)



Electric field and electrons in an LWFA (see [1]) simulation.

The problem of dephasing

Flying focus lasers – solving the problem of dephasing
1. TWEAC

- **T**raveling-**W**ave **E**lectron **AC**celerator
- Developed by Debus et al [2]
- Uses two laser pulses with tilted pulse fronts
- The tilt controls the velocity of the overlapping region

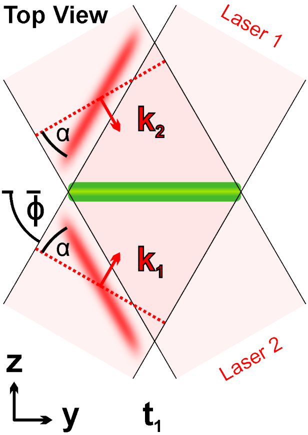


Image: TWEAC setup using two laser pulses. Image taken from Debus [3]

Flying focus lasers – solving the problem of dephasing

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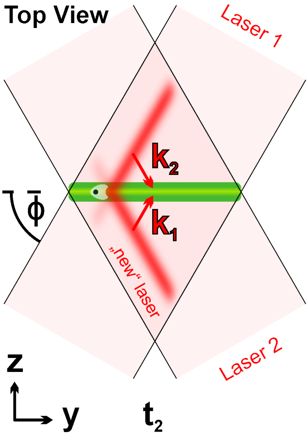

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Flying focus lasers – solving the problem of dephasing
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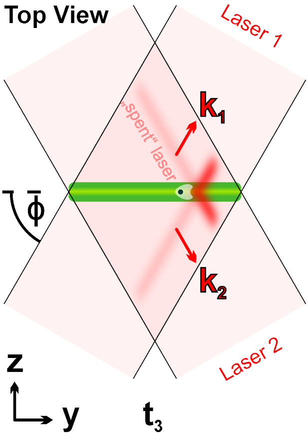
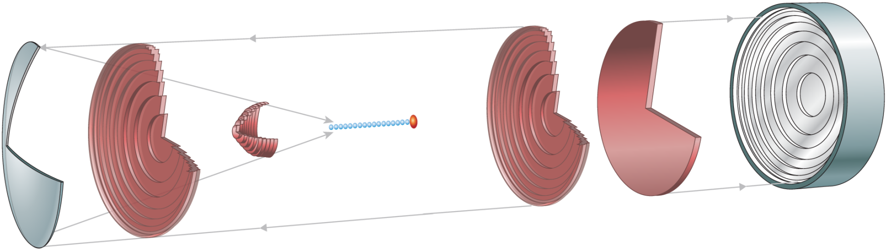


Image: TWEAC setup using two laser pulses. Image taken from Debus [3]

Flying focus lasers – solving the Problem of dephasing

2. Axiparabola laser

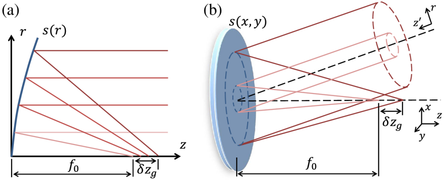


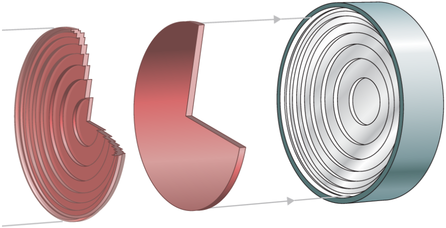
The flying focus setup of Palastro et al [4]. Two optical elements: The Axiparabola (left) and the Radial Group Delay echelon (RGD) (right). Image taken from Palastro et al [4].

Flying focus lasers – solving the Problem of dephasing

2. Axiparabola laser

- Axiparabola [5]:
  - Near-parabolic mirror
  - Focuses light onto a line – the focus region
  - → Light at radius r is focused at $f(r) = f_0 + \delta\left(\frac{r}{R}\right)^2$
- **R**adial **G**roup **D**elay echelon (RGD) [6][4]:
  - Stepped concentric mirror rings
  - Shape follows some function $\tau_D(r)$
  - → controls the timing of the axiparabola focus



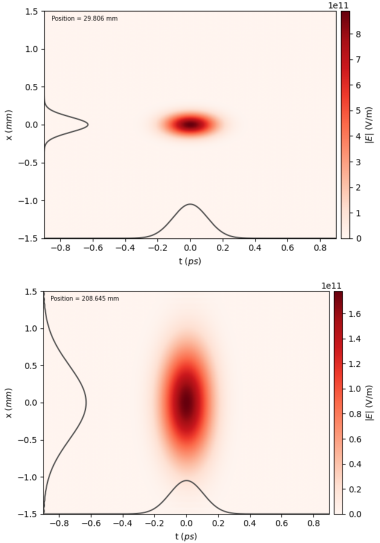


Images: Top: Axiparabola functionality. Image taken from Smartsev et al [5]. Bottom: RGD functionality. Image taken from Palastro et al [4].

2 Flying focus lasers in Lasy and PIConGPU

Lasy
A python library

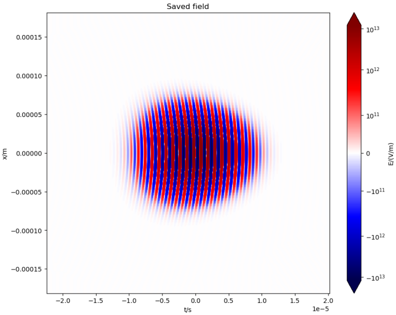
- A python library for simulating Laser pulses in a vacuum [7]
- Uses complex envelope of the laser field
- Uses angular spectrum propagation
- Can use cylindrical coordinates for memory and CPU time efficiency
- Offers a range of optical elements

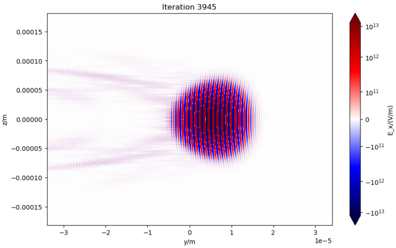


Images: Example of a Gaussian pulse being propagated by Lasy. Top: Generated at the focus. Bottom: 6 z_R after the focus.

Importing to PIConGPU

- New module `full_field`
- Generates full electric field and saves it using `openPMD-api`
- incidentField method FromOpenPMDPulse (see Dietrich [8]) imports the field into PIConGPU



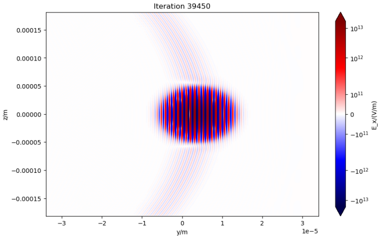


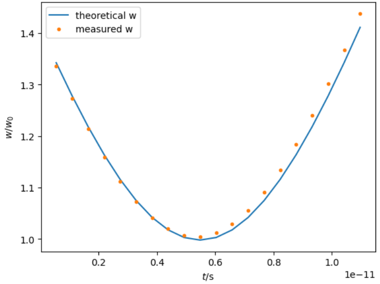
Images: Top: Electric field of a Gaussian laser pulse, $1z_R$ befor the focus, as saved to the file. Bottom: That same electric field entering the simulation window of a PIConGPU simulation.

Testing the method

- Test with Gaussian laser pulse and parabolic mirror
- Some artifacts are visible
  - Problem of the incidentField method

  $\rightarrow$ See [9]
- Test successful



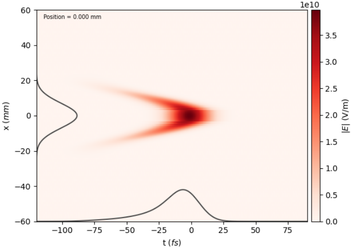


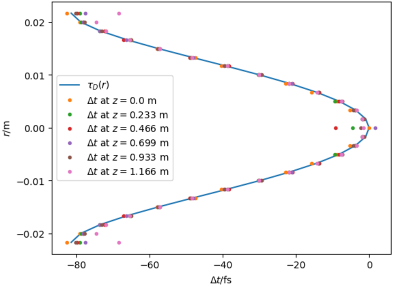
Images: Top: Electric field of the Gaussian laser pulse at the focus according to the PIConGPU simulation. Bottom: The beam waist w of the pulse over time, compared to theory.

TECHNISCHE UNIVERSITÄT DRESDEN DRESDEN concept HZDR

Implementing the flying focus
1. The radial group delay echelon (RGD)

- Implemented from scratch as Lasy optical element
- Following the description by Ambat et al [6]
- Shapes the pulse temporally without focusing or defocusing
- Can generate any radially symmetric shape of delay $\tau_D(r)$

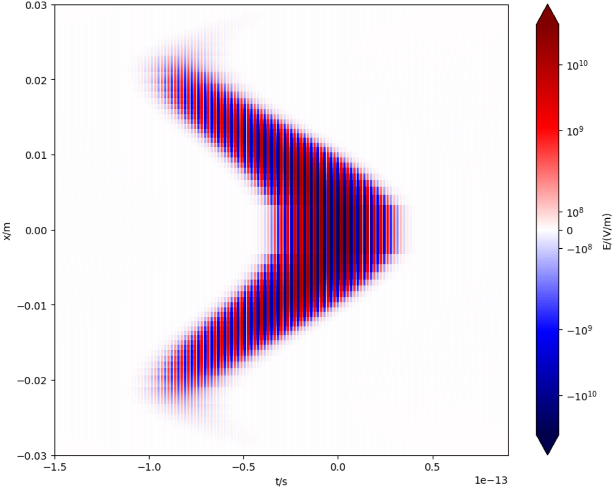




Images: A Gaussian pulse after interacting with the RGD. Top: Field envelope. Bottom: Test results. Even after long distances the shape still holds.

Implementing the flying focus

1. The radial group delay echelon (RGD)

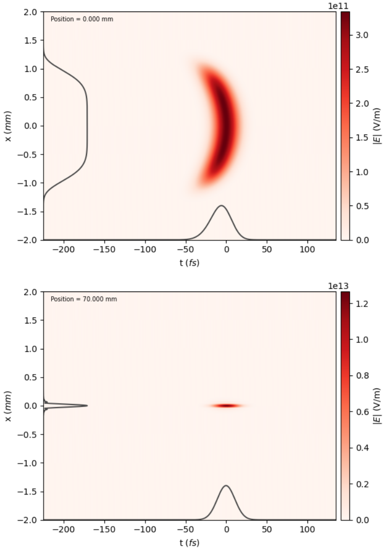


The electric field of the laser after interacting with the RGD.

Implementing the flying focus

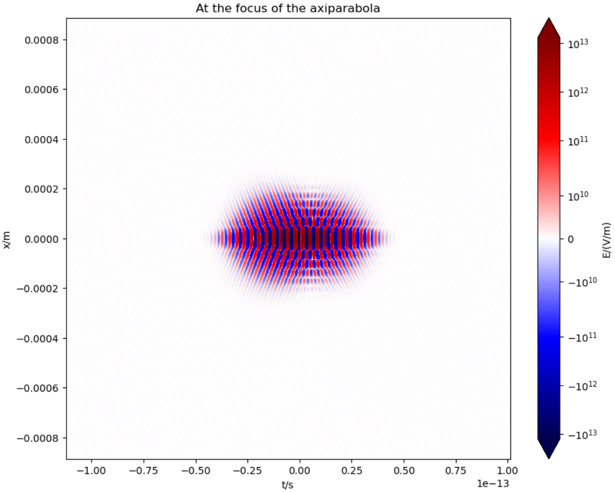
2. The axiparabola

- Included in Lasy
- Following Smartsev et al [5]
- Focusing laser pulse in the focus region
- Also implemented an axiparabola following Ambat et al [6]
  - → Very small differences



Images: A super-Gaussian laser pulse after reflecting off the axiparabola. Top: In the near field. Bottom: In the far field at the beginning of the focus region.

Implementing the flying focus
2. The axiparabola



The electric field of the laser at the beginning of the focus region of the axiparabola.

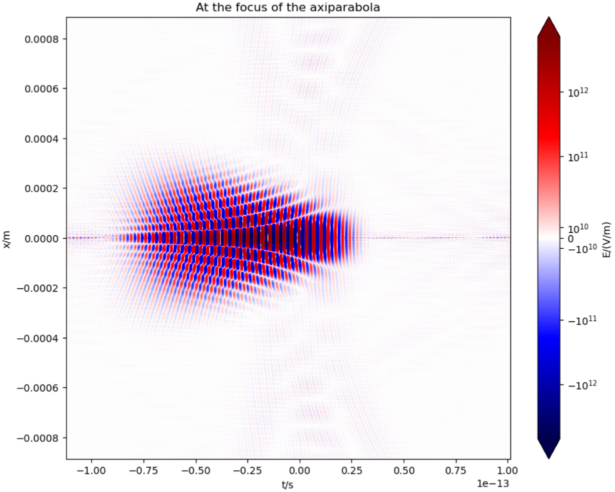
Implementing the flying focus

The complete setup

The simulation had the following steps:

1. Initialising the laser pulse with its pulse profile
2. Applying the RGD to the laser
3. Applying the axiparabola to the laser
4. Propagating the laser to the focus region
   - At this point the laser field can be saved to file
   - The next steps can happen in Lasy or PIConGPU
5. Measure the beam waist w and relative focus position $t_{focus} - f_0/c$ or $z_{focus} - t \cdot c$
6. Propagate the laser some distance through the focus region
7. Repeat steps 5 and 6 until the end of the focus region is reached

Implementing the flying focus
The complete setup



The electric field of the laser at the beginning of the focus region of the complete setup.

Implementing the flying focus
The measurement

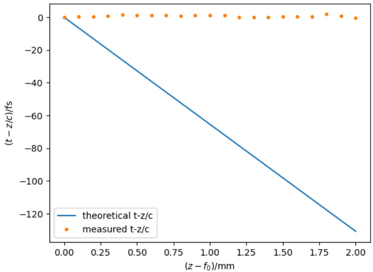
- Beam waist w:
  - ?
- Relative focus position $t_{focus} - f_0/c$ or $z_{focus} - t \cdot c$:
  - ?
- All measurements taken statistically

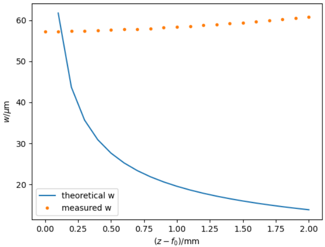
3 Testing the flying focus lasers

Testing the flying focus laser
First results

- Simulation in Lasy
- Result: The flying focus effect cannot be seen

$\rightarrow$ Test failed





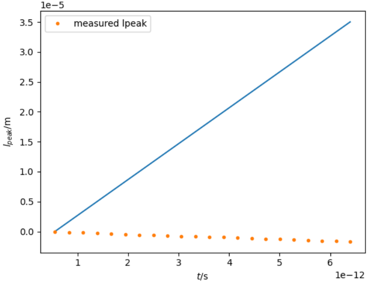
Images: Comparing measurements from the simulations to the expected evolution. Top: From a Lasy simulation: $t - z/c$, Bottom: .

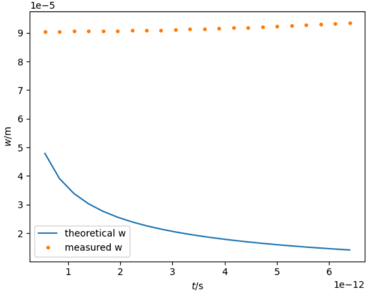
Testing the flying focus laser

The same simulation in PIConGPU

- Simulation in Lasy to the focus region, from there in PIConGPU
- Input file for PIConGPU saved at the beginning of the focus region
- Result: The flying focus effect cannot be seen here either

→ Test failed





Images: Comparing measurements from the simulations to the expected evolution. Top: , Bottom: From a PIConGPU simulation: $z - t \cdot c$.

Testing the flying focus laser
Checking the shapes of the optical elements

- Multiple differential equations for the shape of the RGD $\tau_D(r)$
  - All result in very similar shapes
- Multiple equations for the shape of the axiparabola $s_f(r)$
  - All follow from the same differential equation (and solve it)
  - All are very similar shapes

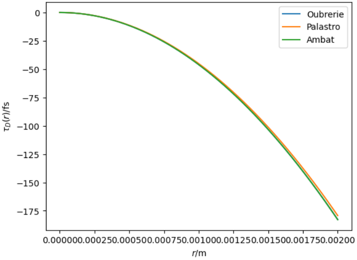


Image: The delay the RGD should achieve according to different sources. The shape of the RGD is similar to this, just stepped in $\lambda_0/2$ steps. Relative differences $\approx 1/50$.

Testing the flying focus laser

Checking the shapes of the optical elements

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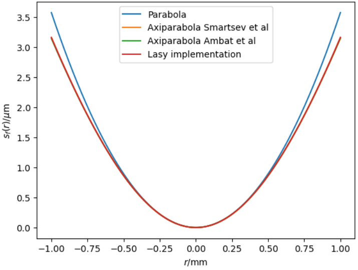
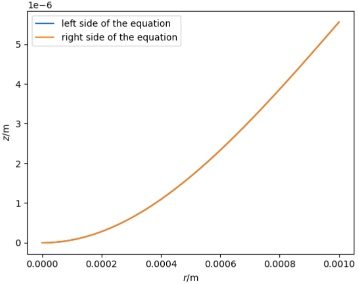




Images: Top: The shape of the axiparabola according to different sources. Relative differences $\approx 10^{-2}$. Bottom: The two sides of the differential equation the axiparabola is derived from. Relative differences $\approx 10^{-5}$.

Testing the flying focus laser

Testing all the used modules

-

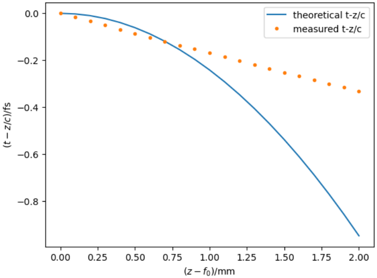
Images:

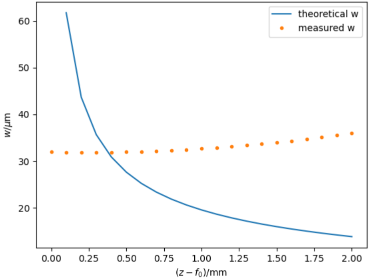
Testing the flying focus laser

Axiparabola only

- With only the axiparabola differences appear already
  - → Here must be the problem
- Using the axiparabola described by Ambat et al [6] gives similar results

→ Contacted the Lasy developers to no avail (so far)





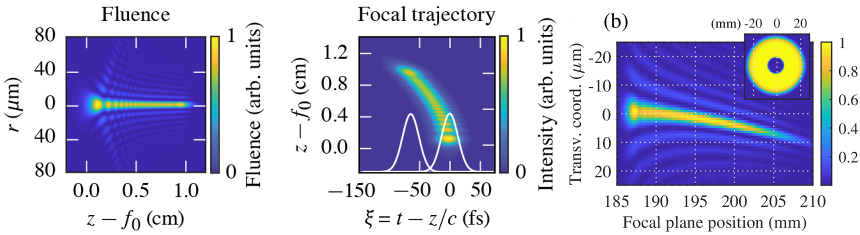
Images: Comparing measurements from a Lasy simulation with theoretical values using Ambat et al [6]. Top: Arrival time $t - z/c$. Bottom: Beam waist w.

4 Summary and Outlook

Summary

These tests were done:

- All the code was tested with Gaussian laser pulses
- The shapes of the optical elements were compared to different sources
- The flying focus pulse was simulated and compared in Lasy and PIConGPU
-
- The Lasy developers were contacted



Simulation results from other papers using only the axiparabola, showing it worked for them without Lasy. Left (Fluence and Focal trajectory): Ambat et al [6]. Right (b): Smartsev et al [5]

Outlook

- Lasy lasers available in PIConGPU

→ LWFA with new laser setups possible

- Some problems still need resolving
- Some tests conducted after writing the bachelors thesis:
  - Trying different propagation methods
  - Initialising the pulse with other pulse profiles
  - Using Cartesian coordinates on nodes with lots of memory

`(*) There's more to explore here.`

References (I)

T Tajima and JM Dawson.
Laser electron-accelerator.
Physical Review Letters, 43(4), 1979.

Alexander Debus, Richard Pausch, Axel Huebl, Klaus Steiniger, Rene Widera, Thomas E. Cowan, Ulrich Schramm, and Michael Bussmann.
Circumventing the dephasing and depletion limits of laser-wakefield acceleration.
Physical Review X, 9, 2019.

Alexander Debus.
Traveling-wave electron accelerators.
AAC 2022 conference talk: `https://www.aac2022.org/`.

J. P. Palastro, J. L. Shaw, P. Franke, D. Ramsey, T. T. Simpson, and D. H. Froula.
Dephasingless laser wakefield acceleration.
Phys. Rew. Letters, 124, 2020.

References (II)

Slava Smartsev, Clement Caizergues, Kosta Oubrerie, Julien Gautier, Jean-Philippe Goddet, Amar Tafzi, Kim Ta Phuoc, Victor Malka, and Cedric Thaury.
Axiparabola: a long-focal-depth, high-resolution mirror for broadband high-intensity lasers.
Optics Letters, 44, 2019.

M. V. Ambat, J. L. Shaw, J. J. Pigeon, K. G. Miller, T. T. Simpson, D. H. Froula, and J. P. Palastro.
Programmable-trajectory ultrafast flying focus pulses.
Optics Express, 31(19), 2023.

Lasy 0.6.2 documentation.
`https://lasydoc.readthedocs.io/en/latest`.
Accessed october 2025.

Fabia Dietrich.
Implementation and validation of a method for using experimentally measured laser profiles in laser-plasma simulations with picongpu.
Master's thesis, HZDR, TU Dresden, 2024.

References (III)

Klaus Steiniger.
Issue 5269 todo in laser refactoring.
https://github.com/ComputationalRadiationPhysics/picongpu/issues/5269.
Accessed october 2025.

Remaining Possible reasons for failure

- The Axiparabola
  - The shape could still be wrong
- The Propagation
  - Possibly the Lasy propagation gives incorrect/incomplete results
- The Findings in the other papers
  - Maybe the Axiparabola does not work at all
  - Maybe it works in a different manner
- The chosen parameters might not fulfill some important approximation condition